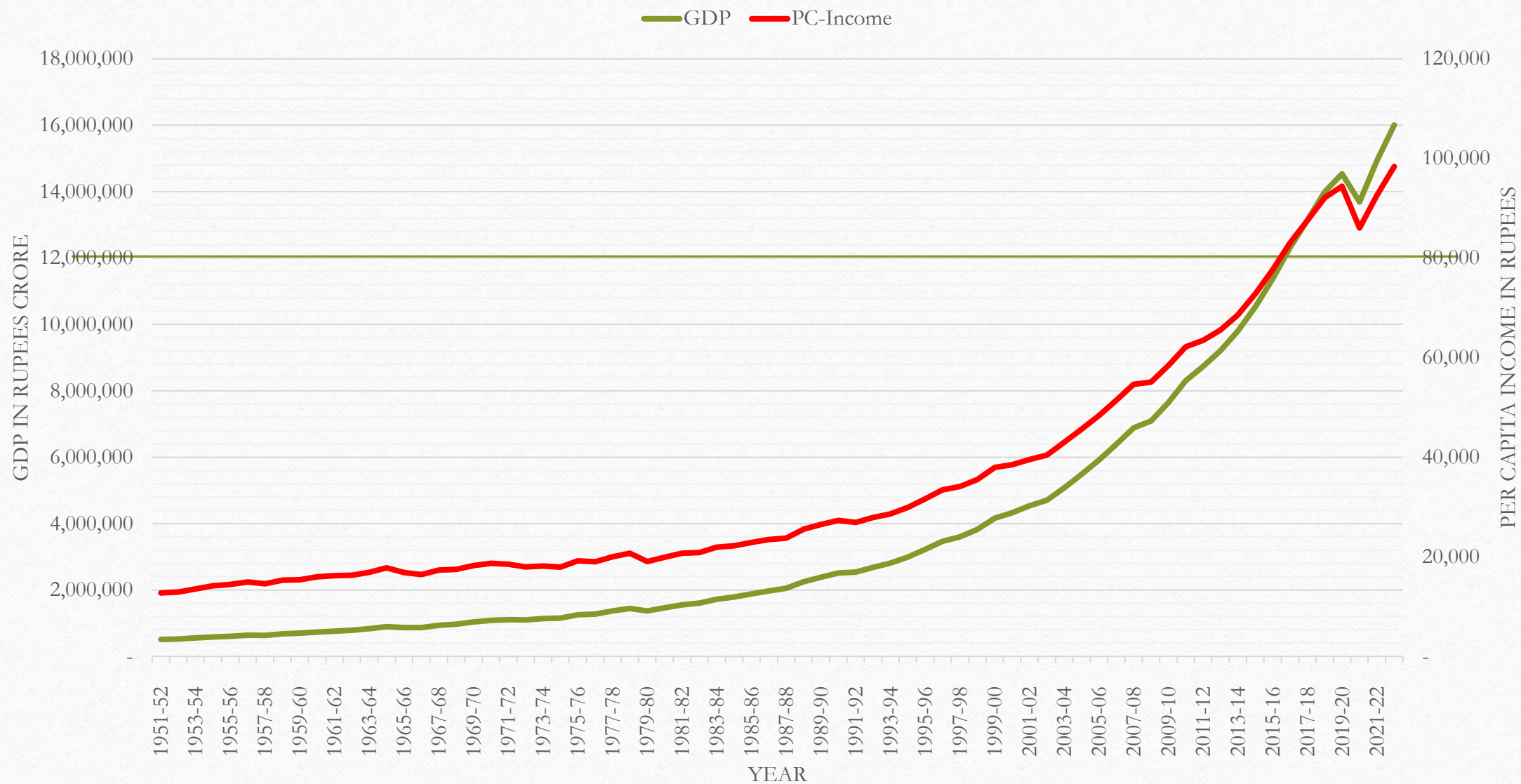


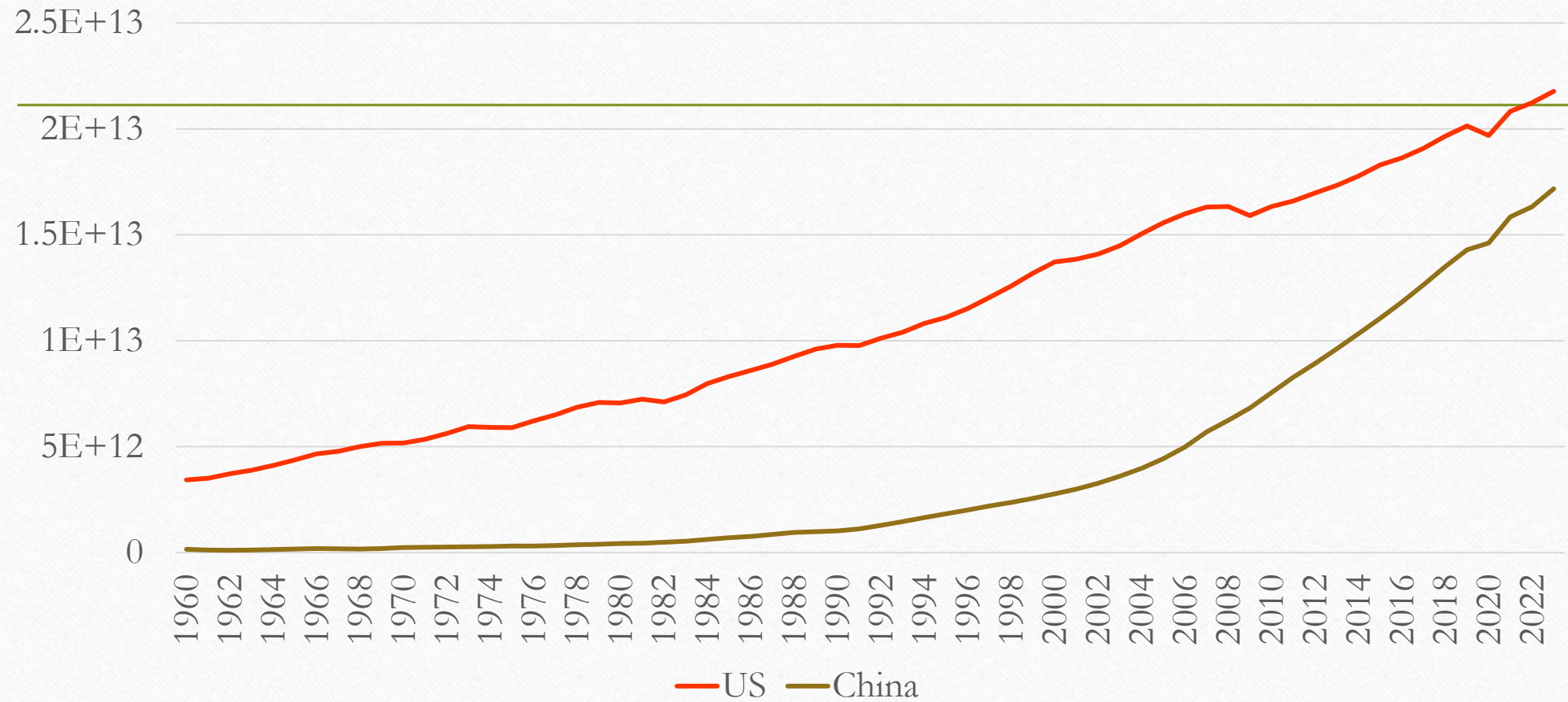


Long-term Macroeconomics & Theories of Economic Growth

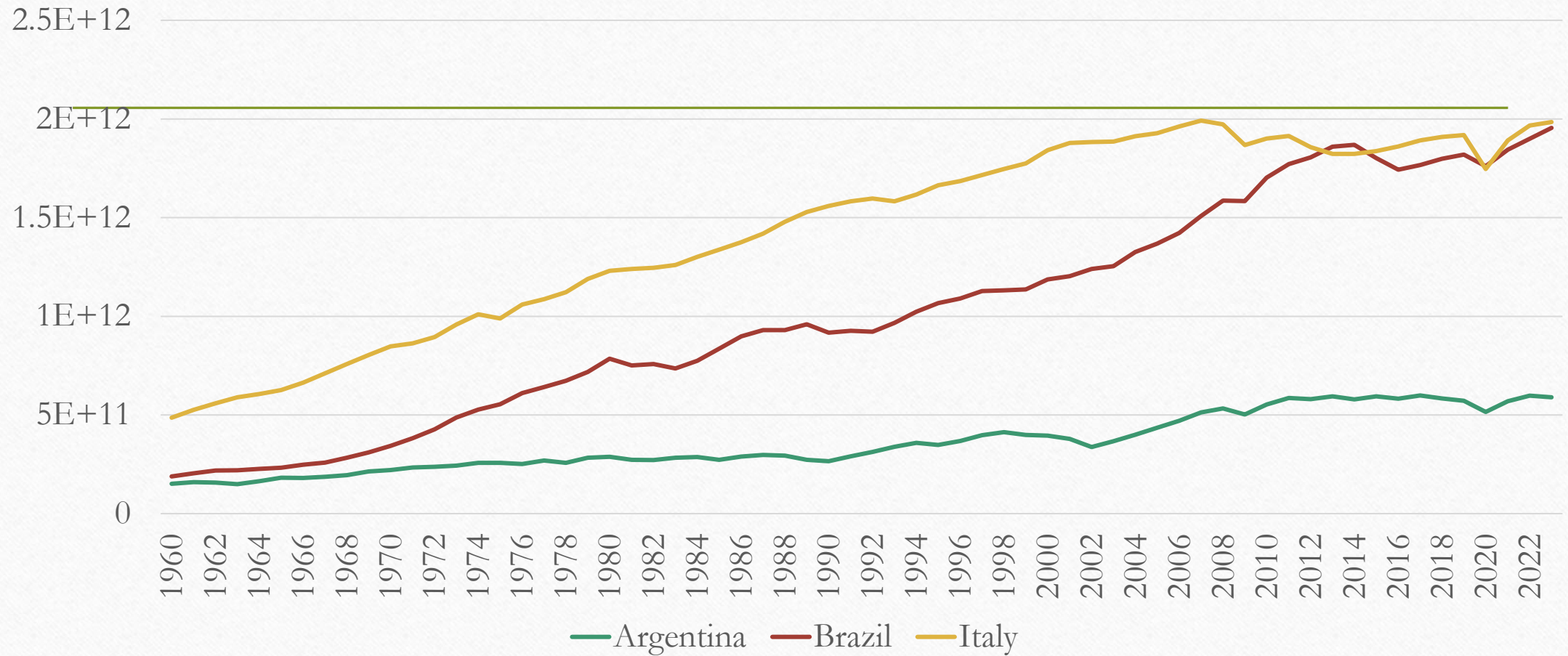
Session 3

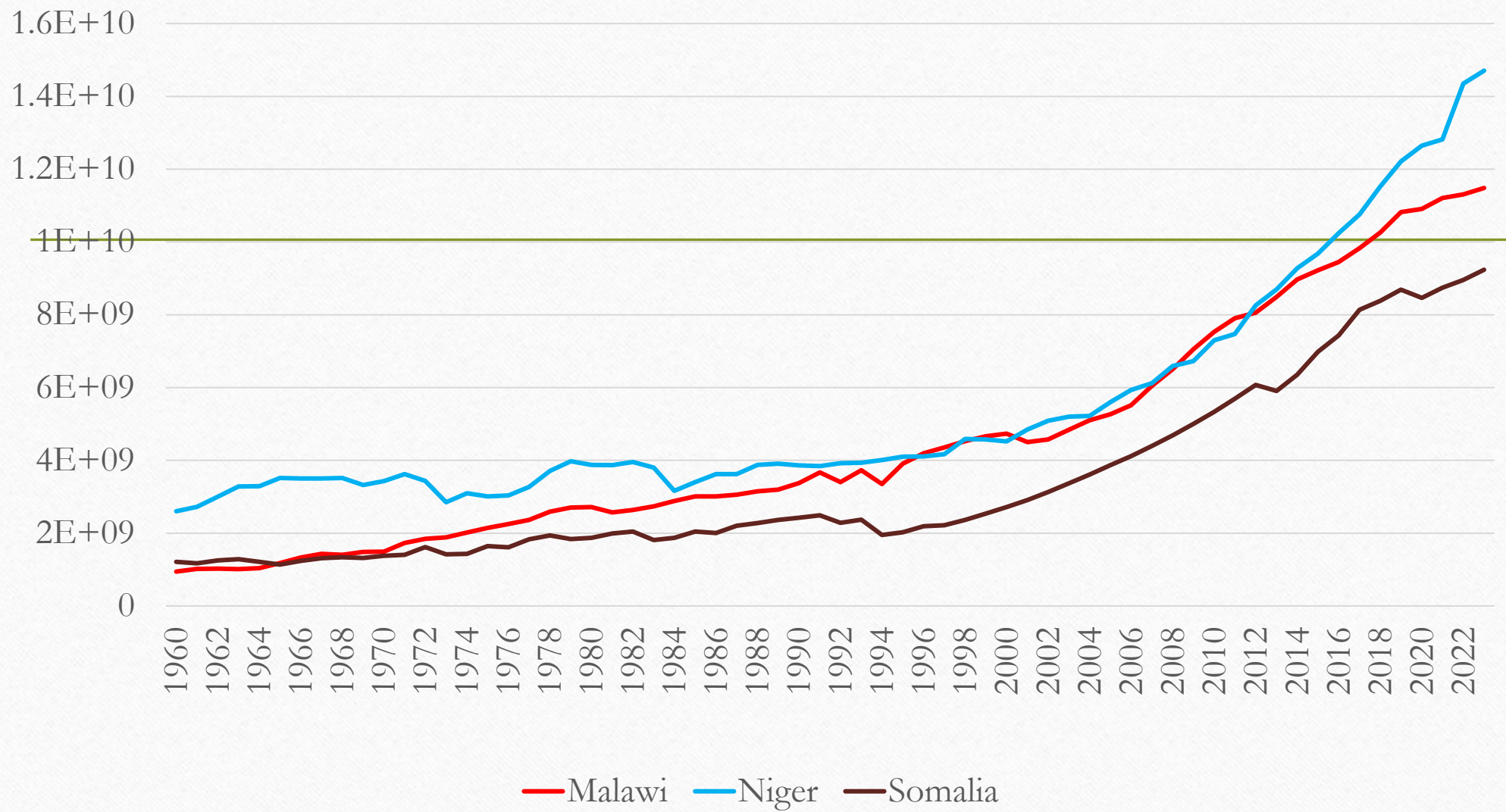


US & China

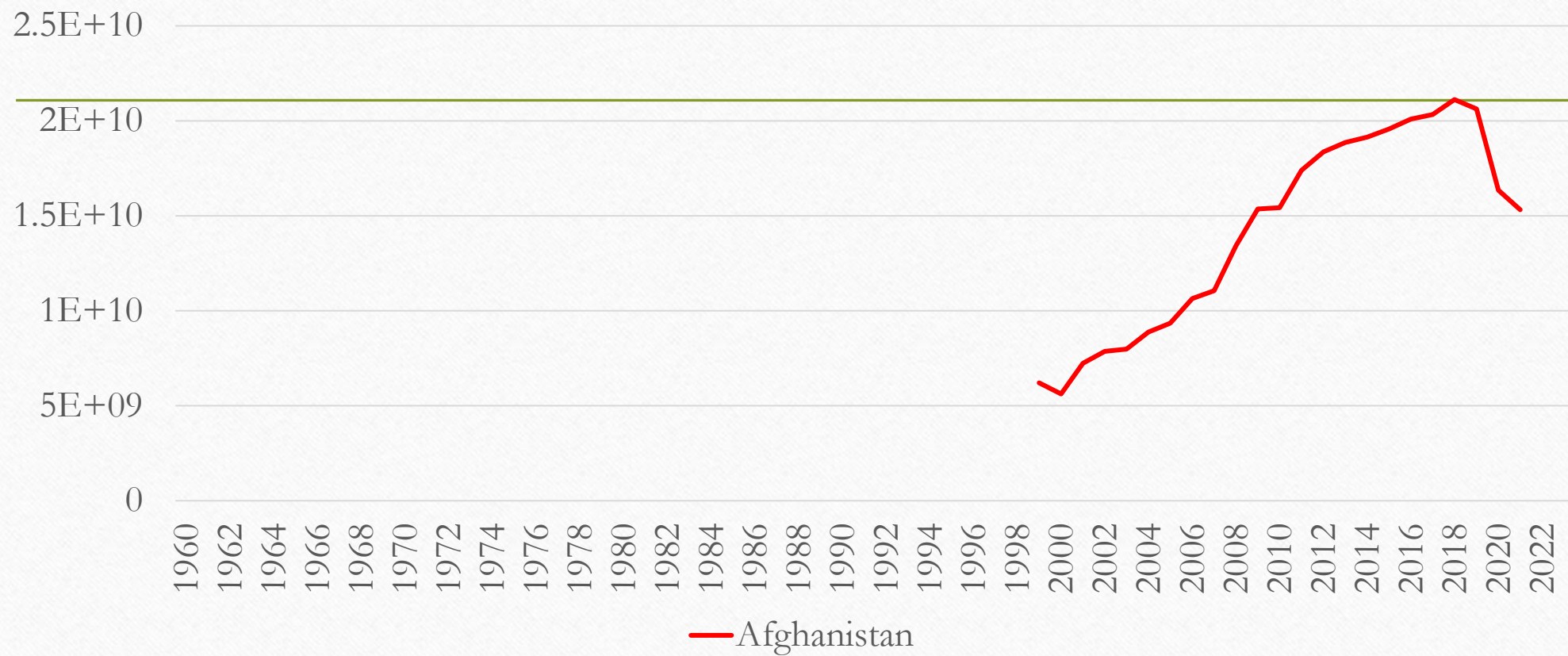


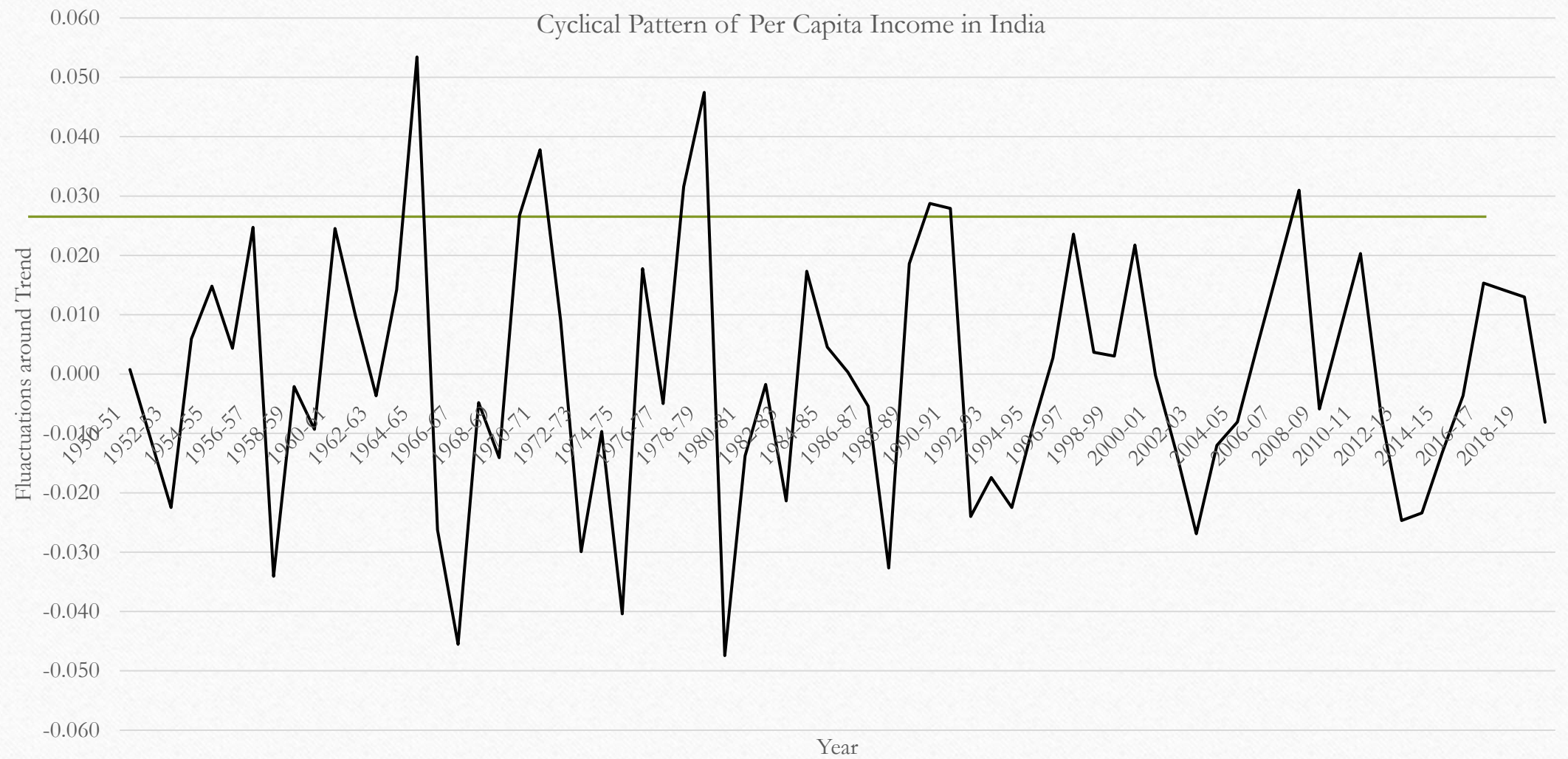
Argentina, Brazil & Italy





Afghanistan





Economic Growth (3 Year MA)

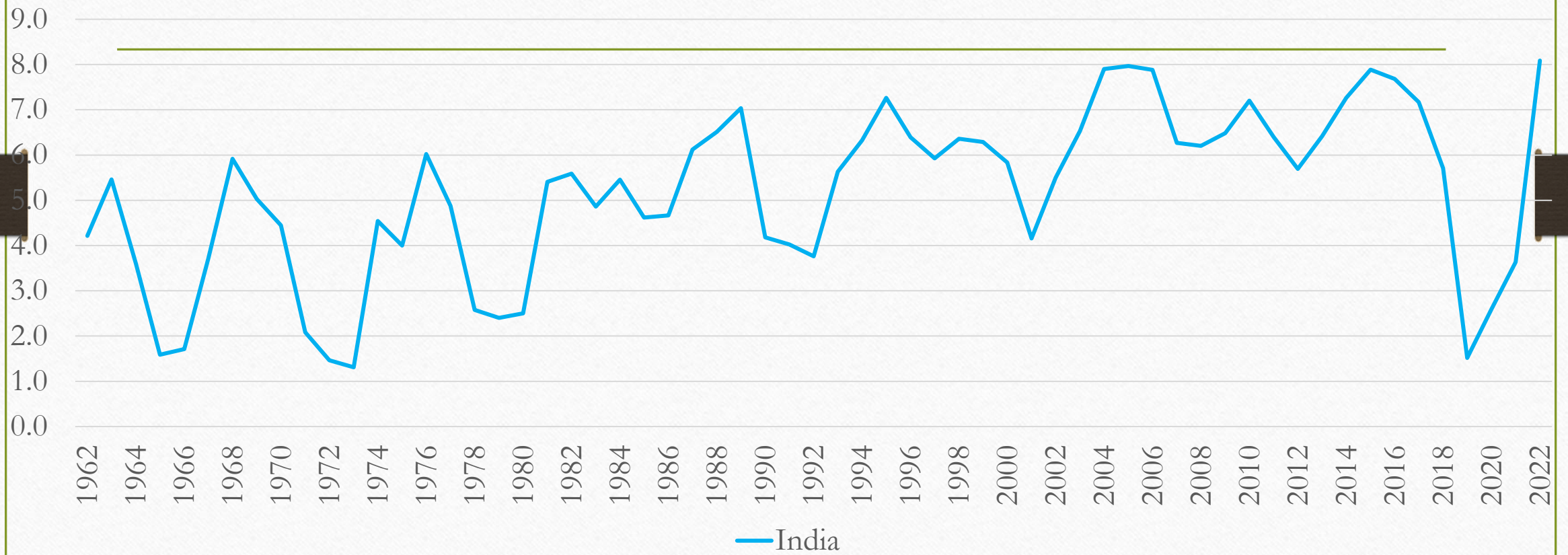
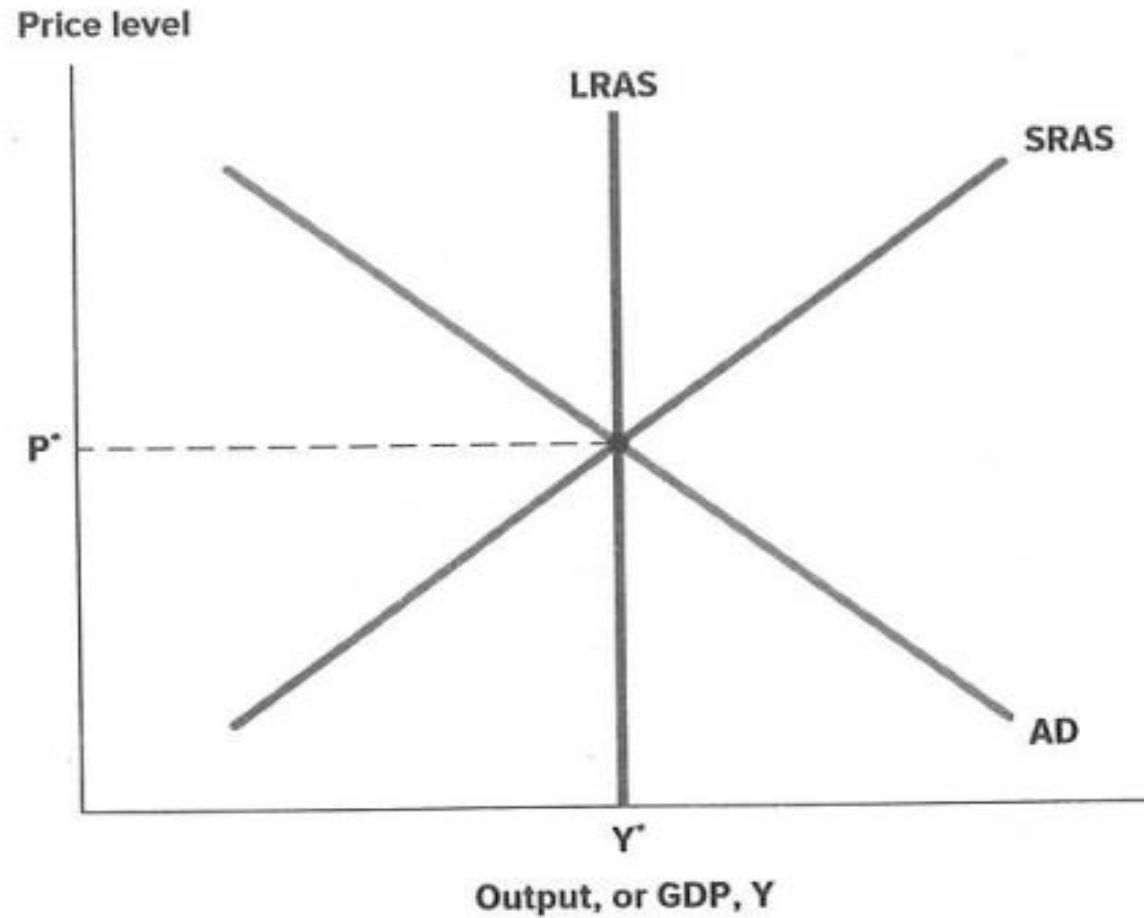


FIGURE 12-9
A model of the
macroeconomy



When the aggregate demand and supply curves are put together, we get a model of the entire economy. The intersection of these curves is the equilibrium, which represents a stable level of prices and output.

Outline



Long-term
Macroeconomic Issues

Theories of Economic
Growth

Problems of Jobless
growth

Long-term Macroeconomic Issues

Subtitle

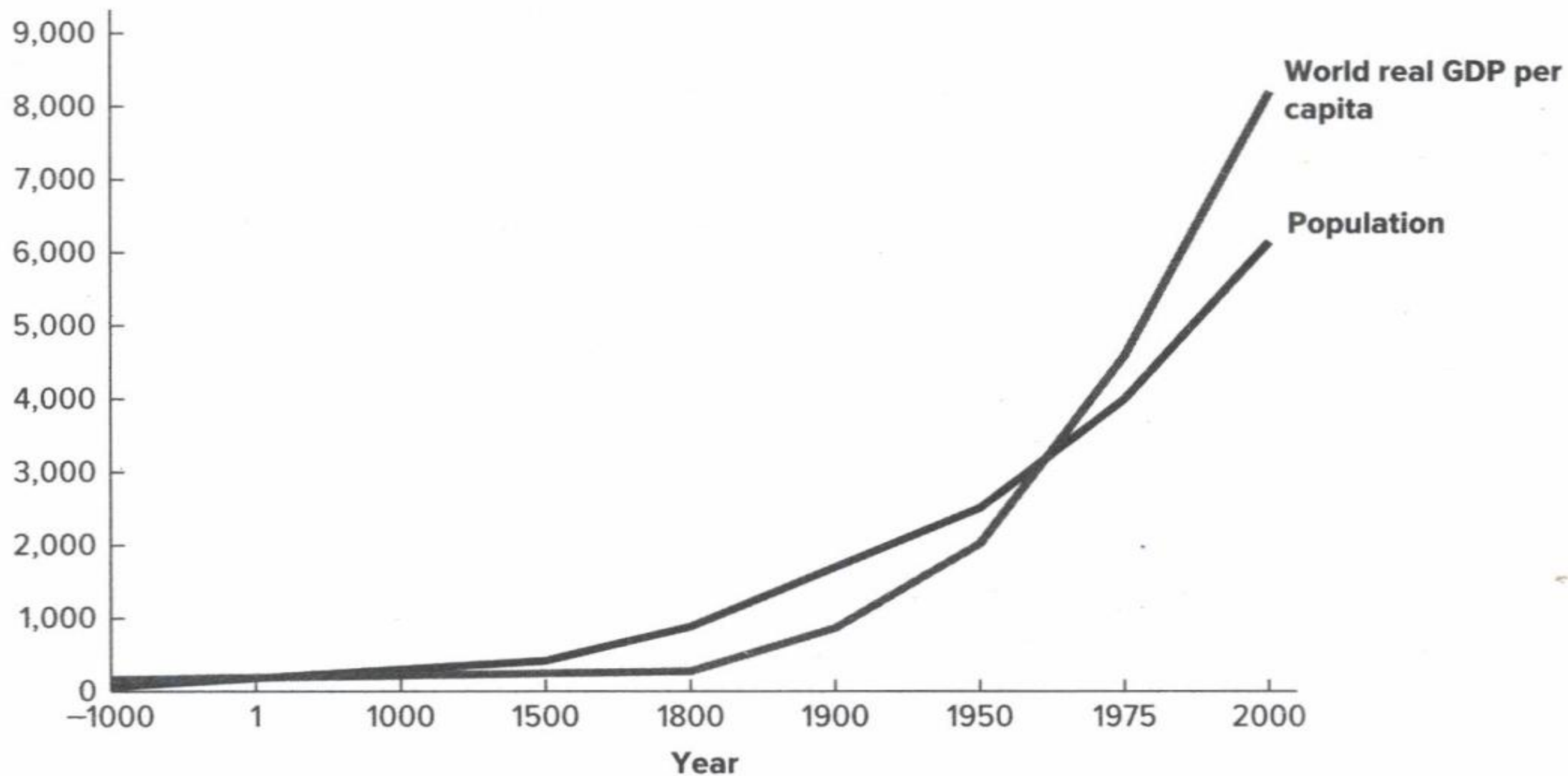


From stagnation to sustained growth

For most of humankind's history, living standards did not change considerably from one generation to the next, despite sporadic important discoveries. These sometimes led to improved quality of life, but growth always stopped eventually.

This was fundamentally changed by the Industrial Revolution, which occurred a little more than two centuries ago. Starting in Britain, and then progressing to other countries, technological innovation and scientific progress resulted in a never-ending cycle of innovation and progress, rather than isolated events. This led to sustained and remarkably stable growth.

Millions of people/
constant U.S. dollars



Last 3000 Years: World Population and Income Per Person

1

- General sense of prices and standard of living
- Historical records and archeological evidences

2

- First 2800 years – Population was growing slowly
- Economy just kept pace with population growth

3

- Last 200 years – global economy underwent radical transformation
- Population grew at a faster pace – growth in per capita GDP outpaced

Industrial revolution and mechanization



Use of machines in the production process

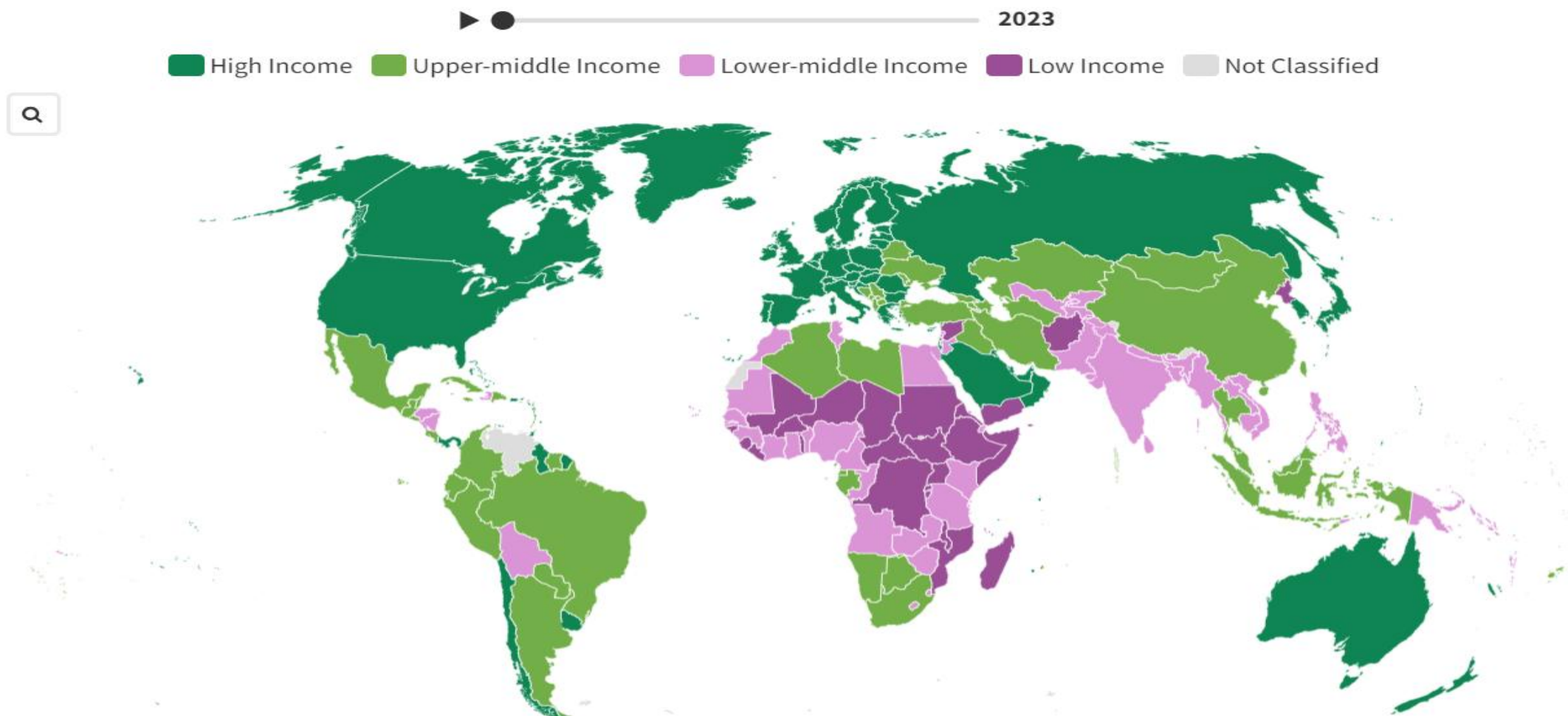
Factories replaced domestic industry

Multifold increase in Labour productivity

Significant improvement in the standard of living

<https://blogs.worldbank.org/en/opendata/world-bank-country-classifications-by-income-level-for-2024-2025>

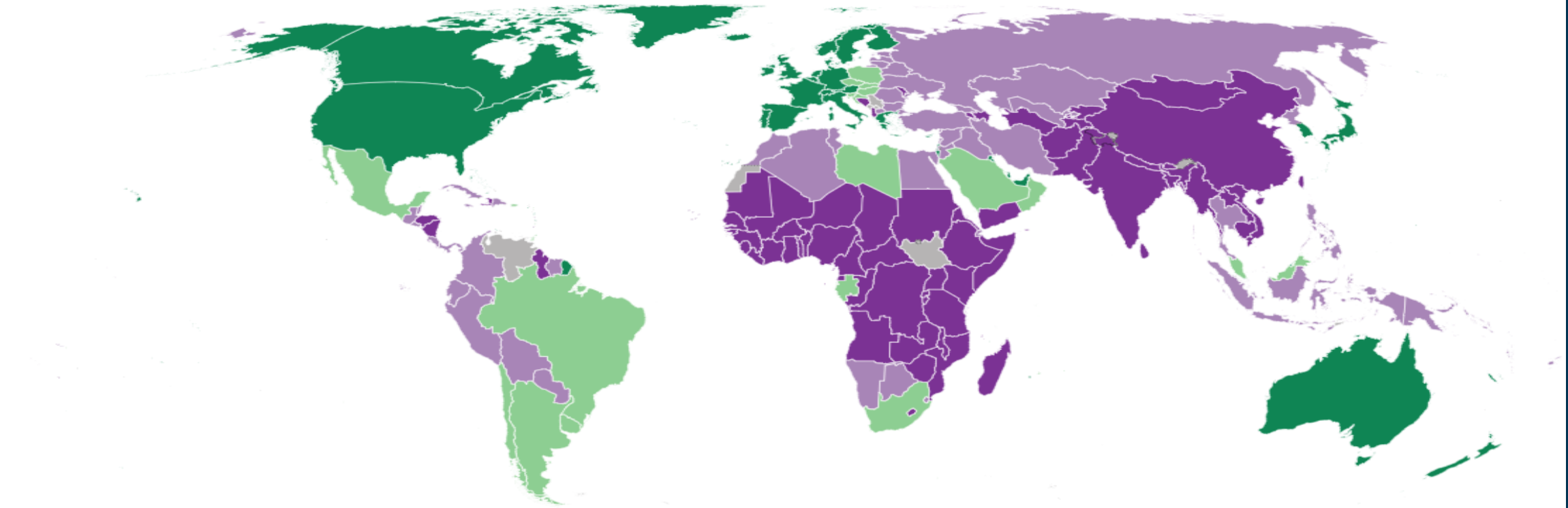
World Bank Group country classification by income level



The world by income

► ● 1996

Low income Lower middle income Upper middle income High income



Source: [World Bank](#)

Note: (1) Countries are classified each year on July 1, the start of the World Bank fiscal year, based on GNI per capita data (World Bank Atlas method) for the previous calendar year. For FY24 the classification uses GNI per capita for 2022. (2) Map boundaries represent boundaries as of 2020 and do not change over time. Country borders or names do not necessarily reflect the World Bank Group's official position. This map is for illustrative purposes and does not imply the expression of any opinion on the part of the World Bank, concerning the legal status of any country or territory or concerning the delimitation of frontiers or boundaries.

Importance of Growth

- Growth rate and compounding
 - A growth rate of 2% will take over 35 years to double GDP
 - A growth rate of 3.5% will take around 20 years for the same
 - Rule of 70

$$\text{GDP}_{1911} = Y + (0.02 \times Y) = 1.02Y$$

$$\text{GDP}_{1912} = 1.02 \times \text{GDP}_{1911} = 1.02 \times 1.02Y = (1.02)^2 Y$$

$$\text{GDP}_{1913} = 1.02 \times \text{GDP}_{1912} = 1.02 \times (1.02)^2 Y = (1.02)^3 Y$$

Importance of Growth

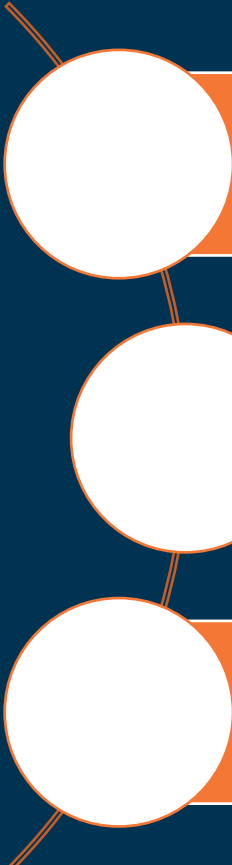
$$\text{GDP}_{\text{yearA}} = \text{GDP}_{\text{yearB}} \times (1 + \text{Growth rate})^{(\text{yearA} - \text{yearB})}$$

$$\text{GDP per capita}_{2015} = \$5,800 \times (1 + 0.02)^{105} \approx \$46,400$$

Importance of Growth

- Say, Indian per capita GDP was Rs. 14,316 in 1947
- An average annual growth rate of 2% of per capita would have taken Indian per capita GDP to Rs.57,258 in 2017
- With an average growth rate of 3.5%, per capita GDP would have been Rs.1,59,091 in 2017
- The difference = Rs.1,45,000 (1,59,091 over 14,316) – 11 times

Important Questions



Why do the countries witness differences in economic growth?

What are the basic determinants of the economic growth?

Can public policy initiatives affect economic growth

Productivity & Economic Growth

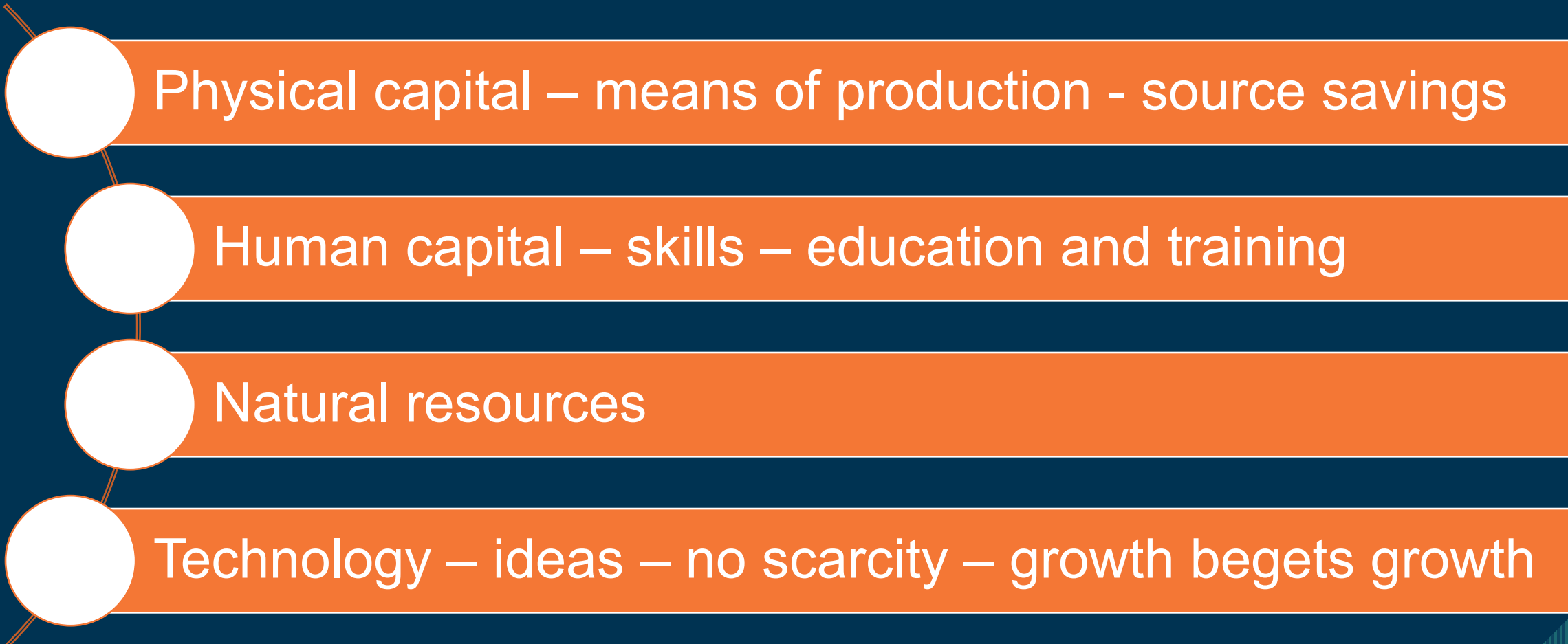
Output
per
Person
(GDP
per
capita)

Higher productivity helps – saving time and efforts in the production process

Saved time and efforts can be used for production of concerned goods as also for others

Higher level of aggregate output can be produced with same amount of labour and capital

Determinants



Physical capital – means of production - source savings

Human capital – skills – education and training

Natural resources

Technology – ideas – no scarcity – growth begets growth

Theories of Growth

Theoretical Framework – Long-run

Theories of Economic Growth

- Explain dynamics of how GDP evolves over time - movement of GDP over a long-term perspective
- Interaction of utility maximizing households and profit maximizing firms determines the long-term pattern of economic activities and evolution of per capita income
- Gradual improvement in thinking – several economists have received Nobel Prize in Economic Sciences – Robert Solow (1987), Nordhaus-Romer (2018), Acemoglu-Johnson-Robinson (2024), Mokyr-Aghion-Howitt (2025) --- Koopmans (1975)

Theoretical Framework

- Household behaviour – Maximising utility out of consumption over infinite number of years of its existence
 - Resources: wage income and returns out of assets
 - Use of Resources: Consumption & Saving
 - Change of assets: Resources over its use
- Firms – Maximise profit by using a production technology which combines factor inputs to produce output
 - Pays wages to workers and return on assets

Interaction of Households & Firms

→ Resources →

Households: Consume & Save

Firms: Hire capital & labour &
Produce

Wages
&
Interest

Production Function (PF)

1

- $Y = A \cdot F(K, L)$
- A technical relation between output and inputs

2

- A: Total Factor Productivity (TFP)
- Improves with technological innovation and conditions for doing business

3

- 'K' stands for capital and 'L' for labour
- More output with accumulation of capital and labour

4

- In per capita terms: $y = A f(k)$

Chapter 1 THE SOLOW GROWTH MODEL

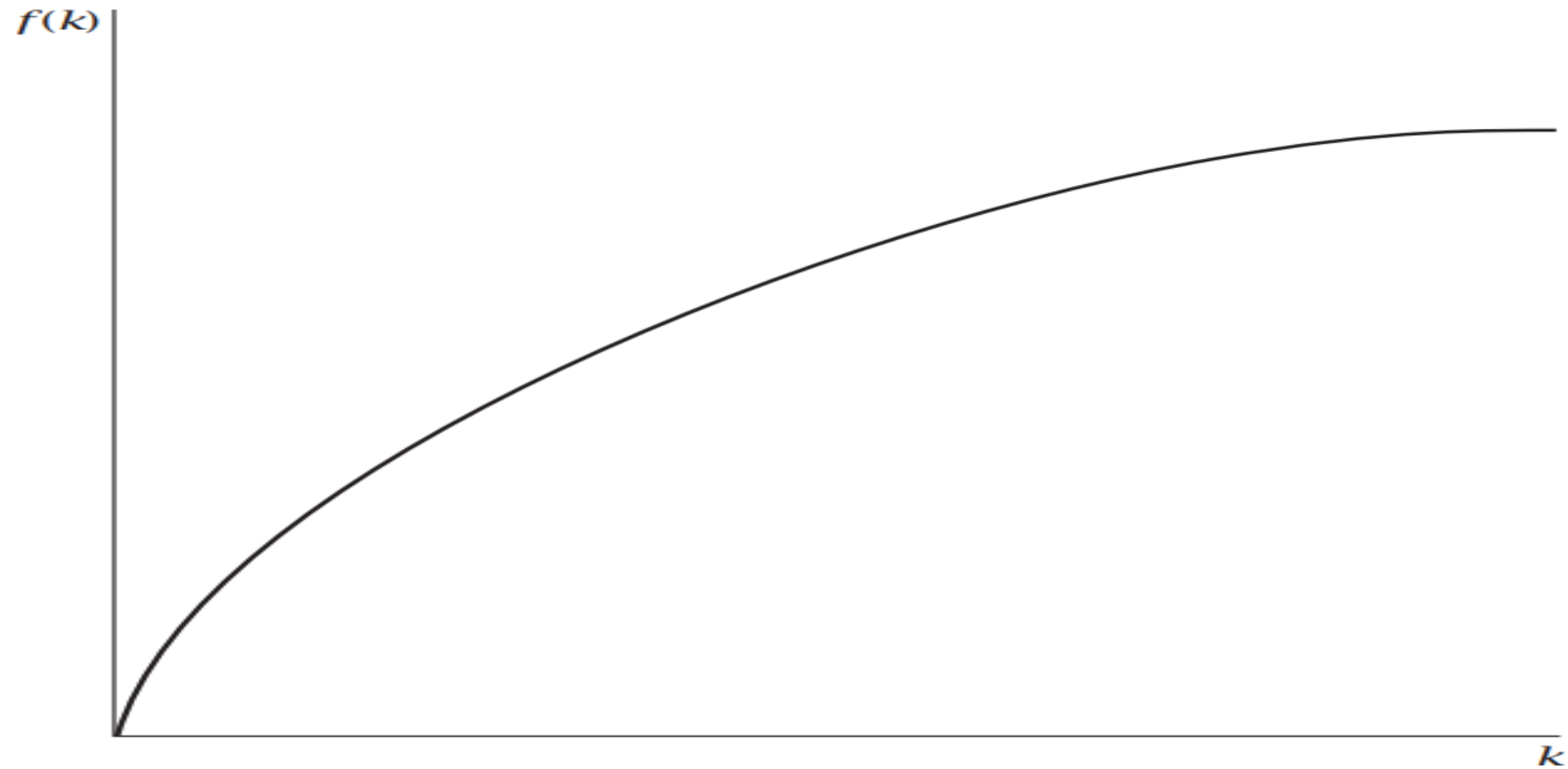


FIGURE 1.1 An example of a production function

Solow Growth Model

Note

PF suggests how 'y' is determined by 'k' (and A)

's' derived from CF determines allocation of output between 'c' and 'i', and 'i' in turn determines 'k'

Two important ingredients in Solow Model

Solow Growth Model

Factors
affecting
'k'

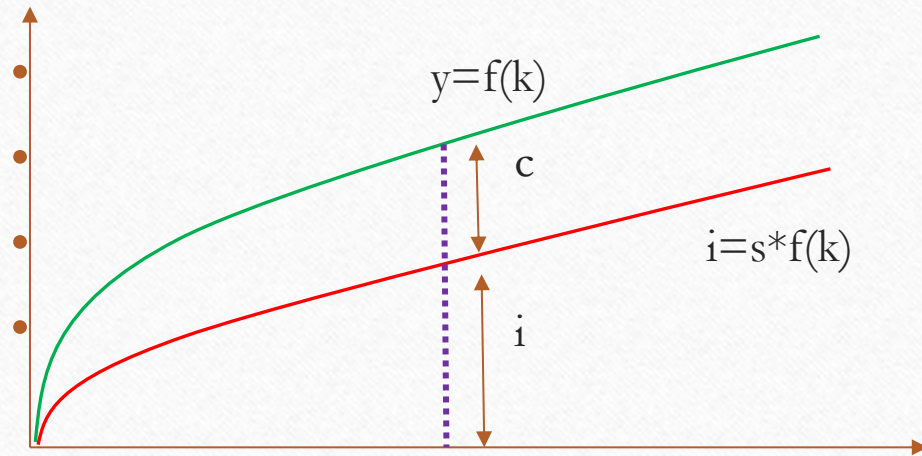
'i' which is addition of machines and equipment contributes to rise in 'k'

$$[i = s \times f(k)]$$

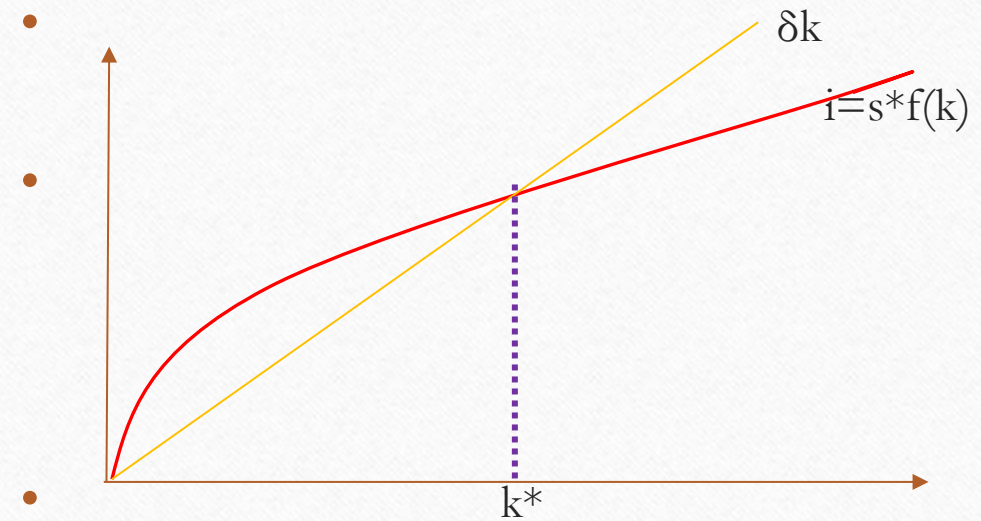
Depreciation of existing assets in the process of being used causes depletion of capital stock $[\delta k]$

Solow Growth Model

'y' and 'i' as a function of 'k'



'i' and depecn as a fn. of 'k'



Solow Growth Model: Steady-State

Rise

- Higher 'k' leads to higher 'y', and higher 'i', as $i = s \times y$
- Rate at which 'y' increases decelerates as 'k' becomes higher

Net rise

- Higher 'k' leads to higher depreciation ($= \delta k$)
- Net change in k ($= \Delta k$) is combination of above two ($= i - \delta k$)

Steady-state

- At 'k*' $i - \delta k = 0$, i.e. $\Delta k = 0$
- This is known as steady-state capital level (capital and output cease to grow)

Solow Growth Model: Transition-State

Long-run
equilibrium
at ' k^* '

If $i > \delta k$, $\Delta k > 0$, so k and y keep on increasing until $k = k^*$

If $i < \delta k$, $\Delta k < 0$, so ' y ' and ' k ' keep on falling until $k = k^*$

Growth miracles in Japan and Germany post-WW II support Solow Model predictions

Key determinants of Economic Growth

- Growth of factor inputs like **workers** and capital provided to the worker
- Saving rate determines the pace of capital accumulation – but given the limits to raise saving rate and diminishing marginal productivity of capital, there is a limit to which they can propel perpetual growth
- Technological progress, which drives productivity growth of physical capital and human capital key to long-term economy prosperity – which is sustained in OECD countries

Determinants of Saving Rate

- Development of the **financial sector**
- Political and **economic** stability
- Tax policy
- Socio-cultural factors

Role of Institutions

THE PRIZE IN ECONOMIC SCIENCES 2024

POPULAR SCIENCE BACKGROUND

The laureates have also developed an innovative theoretical framework that explains why some societies become stuck in a trap with what the laureates call extractive institutions, and why escaping from this trap is so difficult. However, they also show that change is possible and that new institutions can be formed. In some circumstances, a country can break free of its inherited institutions to establish democracy and the rule of law. In the long run, these changes also lead to reduced poverty.

Public Policy Intervention to Improve Technology

- Investment in Education and Health
- Promotion of Science and Technology
- Putting in place and effective implementation of patents, copyrights and intellectual property rights
- Improving overall governance

Accounting for Growth

1

- Explains the contribution of capital, labour and technology - separately
- Captures the relation between growth rate of inputs and the resulting growth rate of output

2

- Growth accounting equation

$$g_Y = g_A + \alpha g_K + (1 - \alpha) g_L$$

α = the share of GDP that is distributed to the owners of capital

$1 - \alpha$ = the share of output that is distributed to labor

Growth Accounting (Solow)

- PF: $Y = A F(K, L)$
- Cobb-Douglas PF: $Y = A K^{\alpha} L^{1-\alpha}$
- $\ln Y = \ln A + \alpha \ln K + (1 - \alpha) \ln L$
- Taking differentiation, both sides, we get
- $dY/Y = dA/A + \alpha (dK/K) + (1 - \alpha) (dL/L)$
- Thus, **economic growth is sum of (i) growth of technology of production, (ii) α times growth of capital, and (iii) $(1 - \alpha)$ times growth of labour**

Accounting for economic growth in the US (Mankiw, 2010)

Years	dY/Y	dA/A	$\alpha (dK/K)$	$(1-\alpha)(dL/L)$
1948-2007	3.6	1.2	1.2	1.2
1948-1972	4.0	1.2	0.9	1.9
1972-1995	3.4	1.3	1.5	0.6
1995-2007	3.5	1.3	1.0	1.3

China

1

- 1978-2004: GDP Growth=9.3%
- Growth of K=10.0%, Growth of L=2.0%
- Share of K in GDP=40%

2

$$0.093 = g_A + (0.4 \times 0.10) + (0.6 \times 0.02)$$

$$g_A = 0.093 - (0.4 \times 0.10) - (0.6 \times 0.02) = 0.041$$

- Technical change accounted for 44% of growth

Special articles

Why India Can Grow at 7 Per Cent a Year or More Projections and Reflections

Based on a simple growth accounting framework, the authors project India's future potential output growth rate through 2025. They argue that there is perhaps more upside potential than downside risks to their central estimate for annual growth, which is close to 7 per cent, or 5.5 per cent per capita.

DANI RODRIK, ARVIND SUBRAMANIAN

India (Rodrick and Subramanian, 2004 - EPW)

- Projected growth for 2004-2025: 6.7%
- Growth in total factor productivity (dA/A): 2.5%
- α times growth in capital: 2.9%
 - $\alpha = 0.35$, and $dK/K = 8.28$
- $(1 - \alpha)$ times growth in labour: 1.3%
 - $1 - \alpha = 0.65$, and $dL/L = 1.9$
- Actual average growth during 2005-06 to 2022-25: 6.7% (Excluding 2020-21 & 2008-09) and 6.5% (Otherwise)

Learning Points

1

- Rapid and sustainable economic growth is critical for augmenting economic welfare in the long-run

2

- Accumulation of capital and employment growth along with improving productivity are key ingredients to sustain growth

3

- At a fundamental level inclusive economic and political institutions are critical

The background features a dark blue field on the right and a light blue field on the left, separated by a diagonal line. A thin, dark blue line runs parallel to the diagonal line, and a thin, light blue line runs parallel to the dark blue line.

Discussion